Task_1 : Introduction to Digital Forensics

1. Basic Forensics Concepts

Digital forensics is a specialized field within forensic science that deals with the identification, preservation, examination, and presentation of digital evidence in a way that is admissible in a court of law. It is commonly applied in the investigation of cybercrimes, data breaches, financial fraud, intellectual property theft, and other digital-related incidents. The key principles of digital forensics include:

• **Integrity** – Ensuring that the digital evidence remains unaltered throughout the investigation process.
• **Authenticity** – Confirming that the evidence is genuine and obtained from a reliable and legitimate source.
• **Repeatability** – Allowing other investigators to achieve the same results by following the same procedures and methods.
• **Legal compliance** – Adhering to relevant laws, regulations, and standard protocols during the investigation.

To preserve data integrity, forensic investigators commonly use hash functions (such as MD5 or SHA-256) to ensure that the data remains consistent and unchanged at every stage of the investigation.

2. Types of Digital Evidence

Digital evidence refers to any data stored or transmitted in digital form that can support an investigation.

a) Volatile Evidence

Volatile data exists temporarily and is lost when a system is powered off.
Examples:

- RAM contents
- Running processes
- Network connections
- Cache memory

This evidence must be collected **first** because it disappears quickly.

b) Non-Volatile Evidence

Non-volatile data is stored permanently and remains even after shutdown.
Examples:

- Hard disks and SSDs
- USB drives
- Memory cards
- System logs

c) Cloud and Remote Evidence

This evidence is stored on remote servers or online platforms.
Examples:

- Emails
- Social media data
- Cloud storage (Google Drive, OneDrive)
- IoT device logs

Each type of evidence requires different collection and analysis techniques.

3. Chain of Custody

The **Chain of Custody** is a documented process that records **who collected, handled, transferred, analyzed, and stored the evidence** from the time it was seized until it is presented in court.
It includes:

- Date and time of evidence collection
- Name and signature of personnel
- Purpose of transfer
- Storage location

If the chain of custody is broken or incomplete, the evidence may be considered **tampered or unreliable**, making it legally inadmissible.

4. Basic Digital Forensic Investigation Process

1. Identification
Recognizing potential sources of digital evidence such as computers, smartphones, servers, or network logs.
2. Preservation
Protecting evidence from alteration using methods like:

- Write blockers
- Forensic imaging
- Secure storage

3. Analysis
Examining the collected data to find relevant information such as:

- Deleted files
- Hidden data
- Metadata
- File timestamps

4. Documentation
Recording every step taken during the investigation, including tools and techniques used.
5. Presentation
Preparing a clear and understandable report for legal authorities, management, or courts.

5. Evidence Handling Techniques

Proper evidence handling ensures reliability and legal validity.
Write Blockers
Hardware or software devices that prevent data from being modified on storage media during analysis.
Hashing
A mathematical process that generates a fixed-length value (hash) for data. Matching hash values before and after analysis confirms data integrity.
Secure Storage
Evidence must be stored in:

- Anti-static bags
- Locked evidence rooms
- Access-controlled environments

6. Documentation Methods

Documentation is critical in digital forensics.
It includes:

- Date and time stamps
- Actions performed

- Tools and versions used
- Investigator details
- Screenshots and photographs

Proper documentation ensures **transparency, accountability, and credibility** of the investigation and follows standards such as **NIST guidelines**.

7. Legal Considerations in Digital Forensics

Digital forensic investigations must comply with national and international laws.
Important legal aspects include:

- **Privacy laws** (e.g., GDPR)
- **Computer misuse laws** (e.g., CFAA)
- **Electronic communication laws** (e.g., ECPA)

Investigators often require **search warrants** to seize devices. Failure to follow legal procedures can lead to evidence being rejected in court.
Ethical concerns also arise when handling personal or sensitive data, making confidentiality essential.

8. Introduction to Digital Forensic Tools

Some commonly used digital forensic tools include:

- **FTK Imager** – Used to create forensic images of disks and storage devices.
- **Autopsy** – An open-source tool for disk image analysis and file recovery.
- **Wireshark** – Used for capturing and analyzing network traffic.
- **RegRipper** – Extracts and analyzes Windows Registry data.

These tools help investigators collect, analyze, and interpret digital evidence efficiently

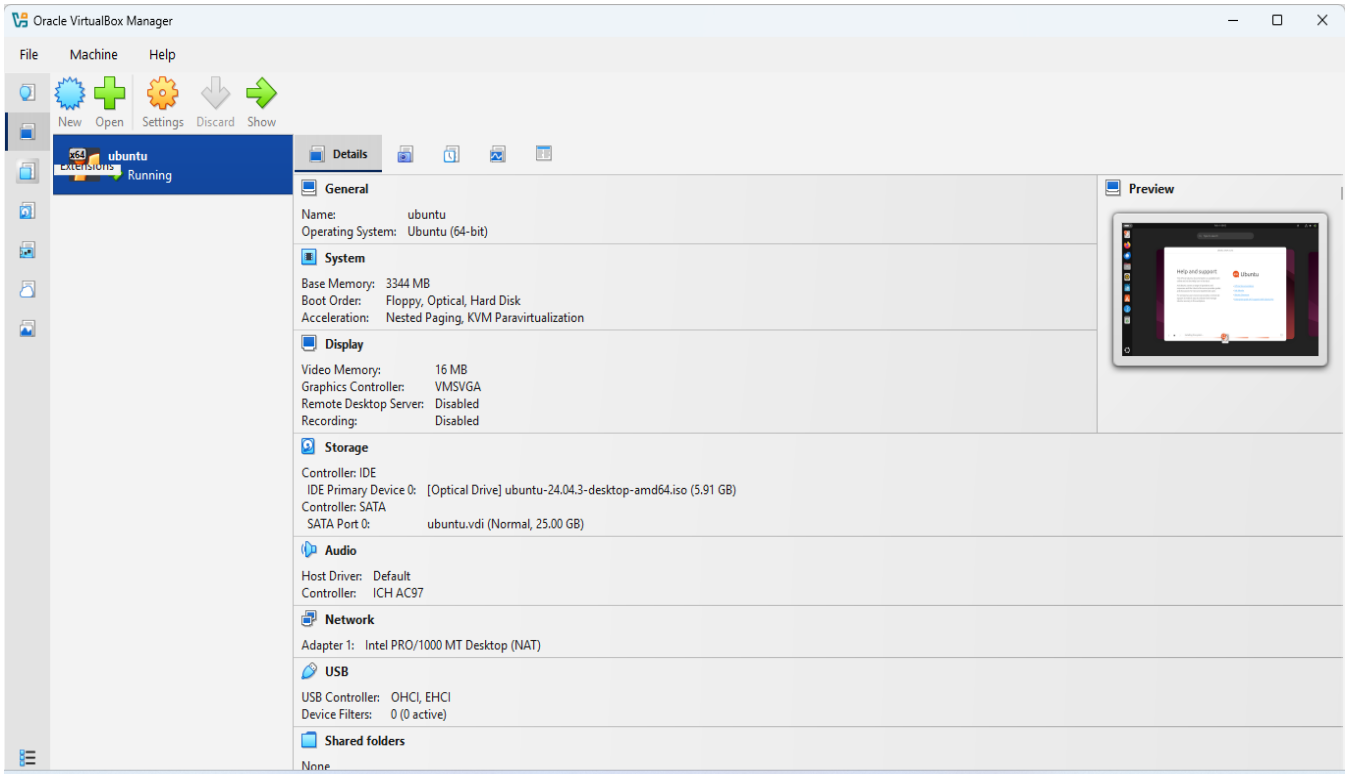
Practical: Basic Forensics Setup

1. Install VirtualBox/VMware Workstation

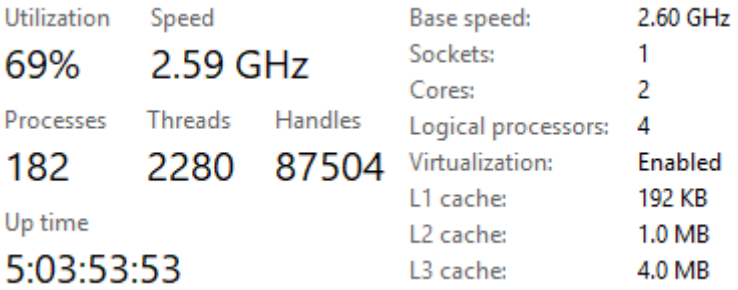
Steps:

- Download from official sites.

- Install and verify compatibility with your OS.



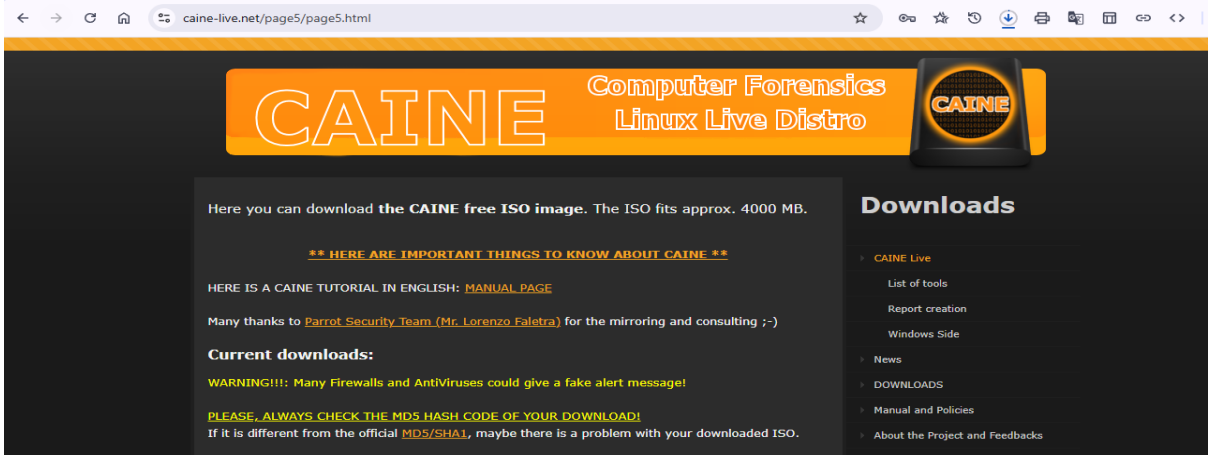
- Why: Isolate forensics tools in a VM to avoid contaminating host systems.



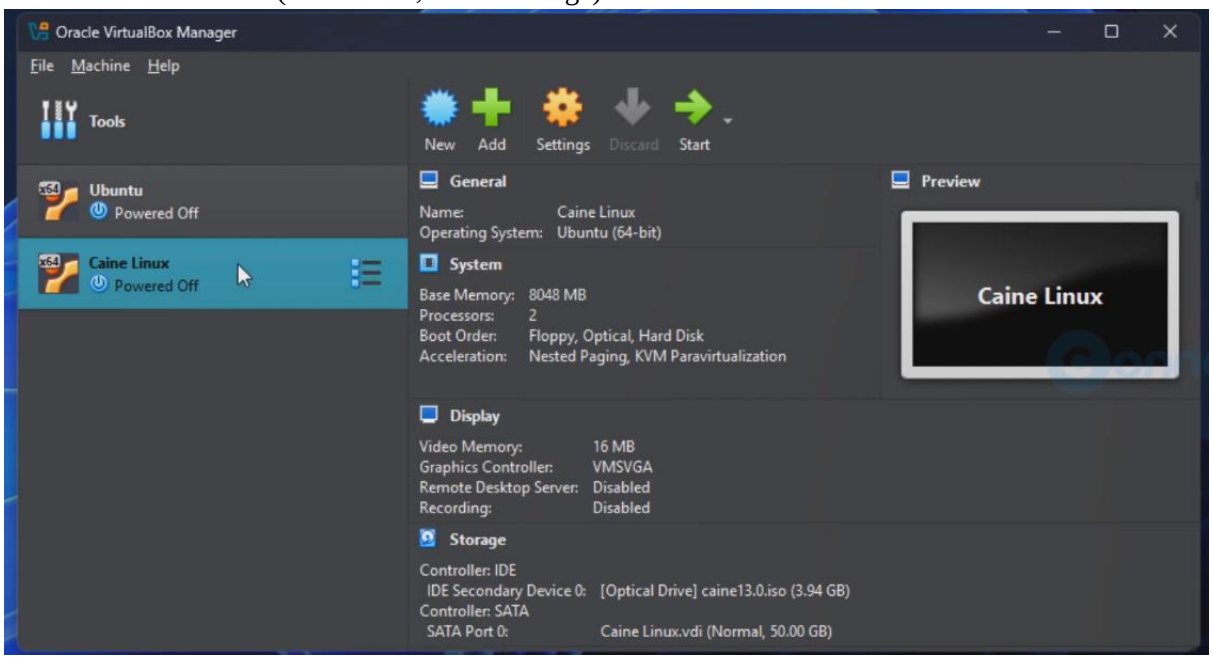
2. Setup Ubuntu/Windows Forensics Workstation

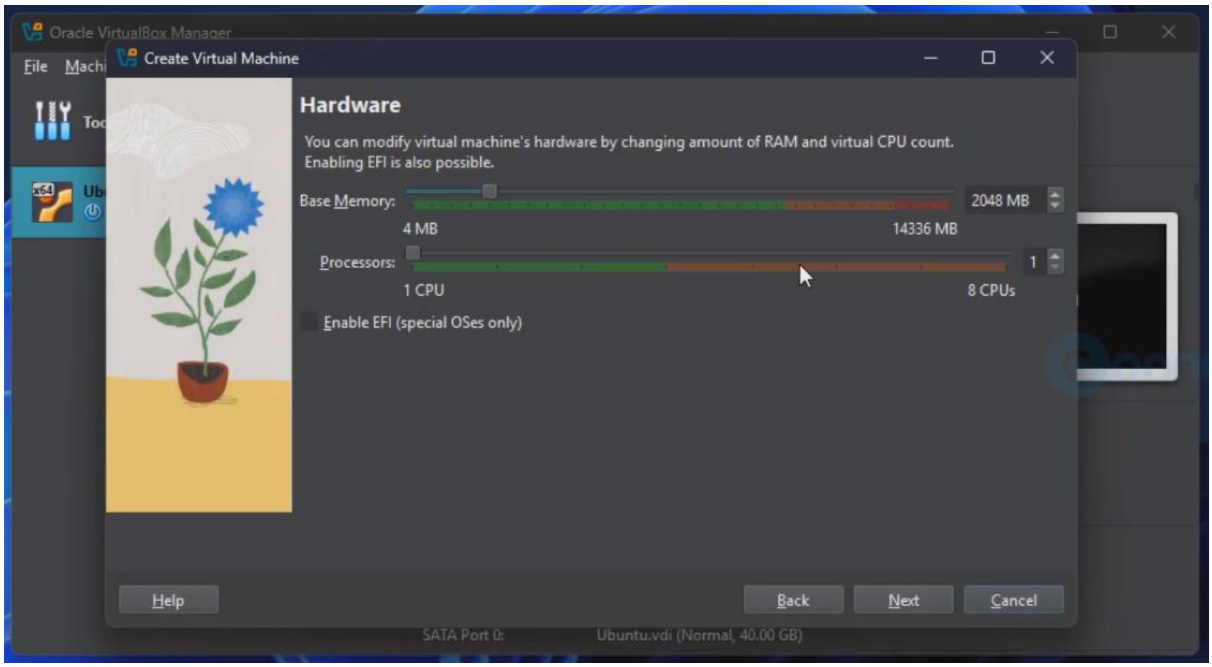
Steps:

- Download a VM image (e.g., Caine Linux for forensics).



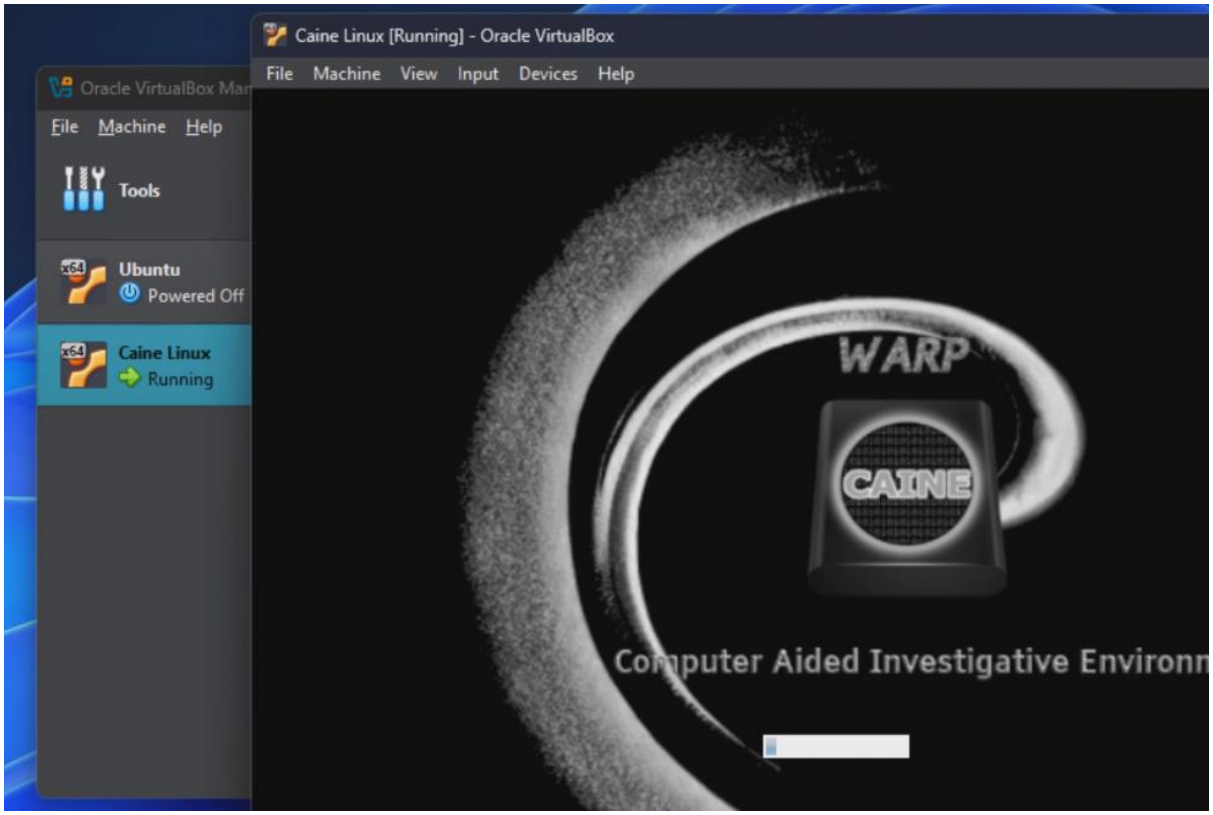
- Allocate resources (4GB RAM, 50GB storage).





- Why: A dedicated OS ensures tool compatibility and reduces external interference.

Setup completed

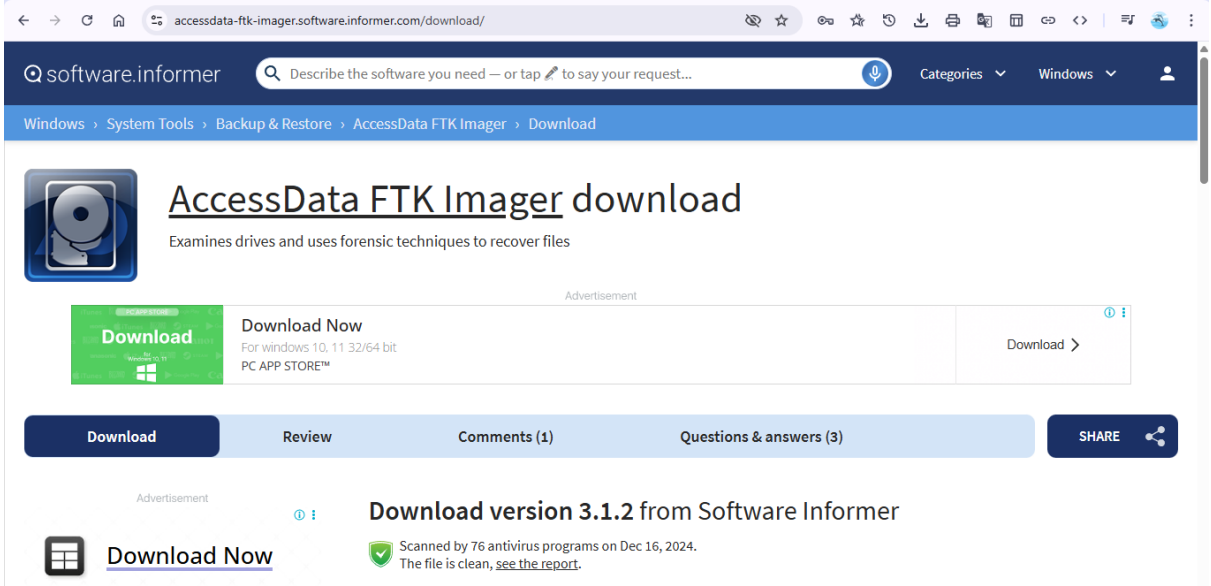


Caine Setup is successfully installed

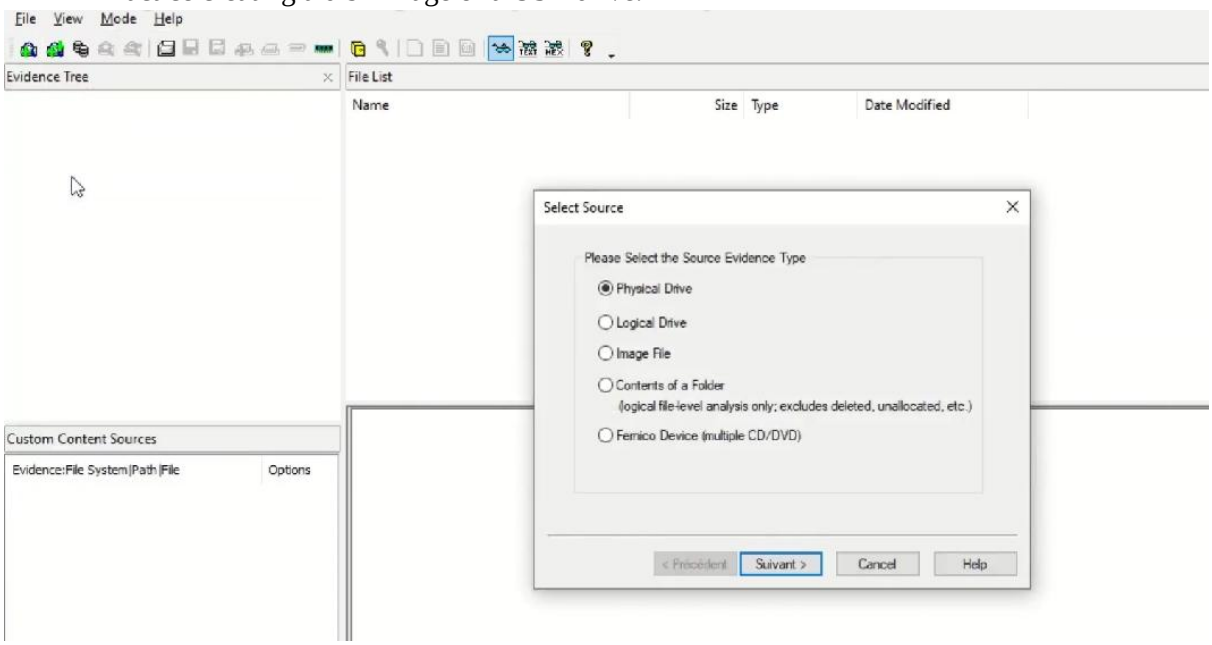
3. Install FTK Imager or FTL Imager Lite

Steps:

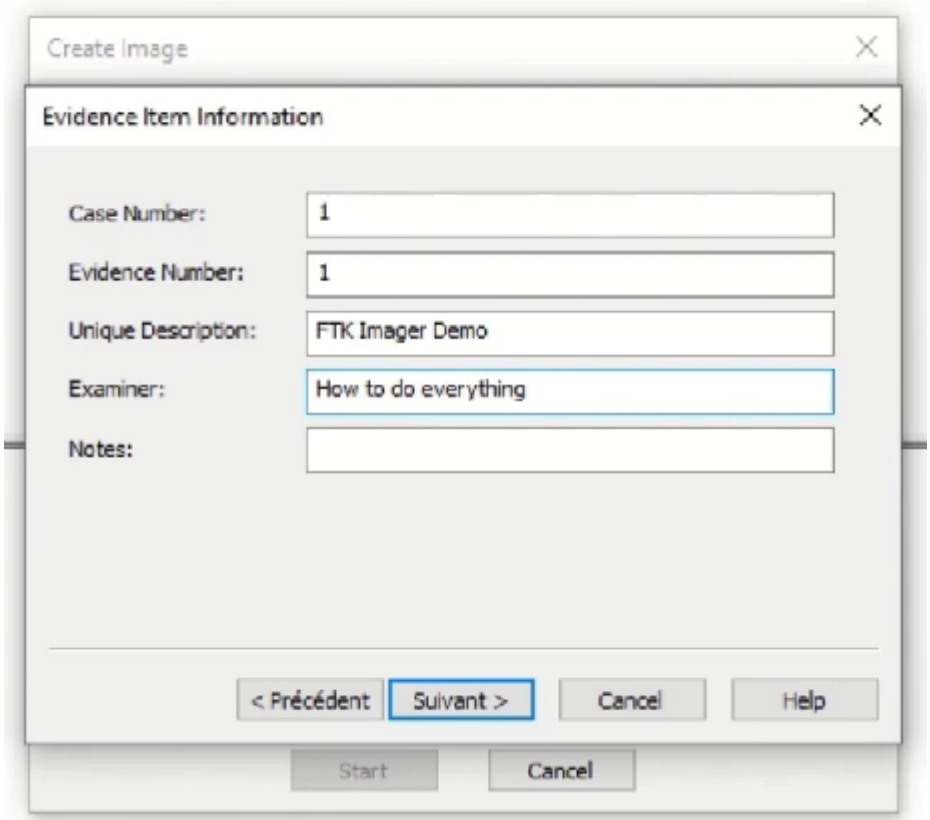
- Download from AccessData.



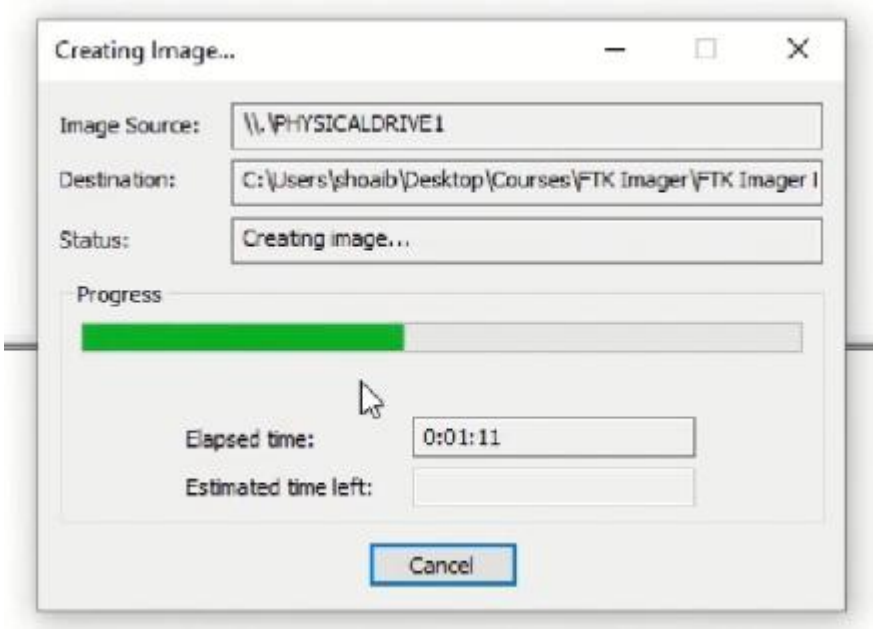
- Practice creating a disk image of a USB drive.



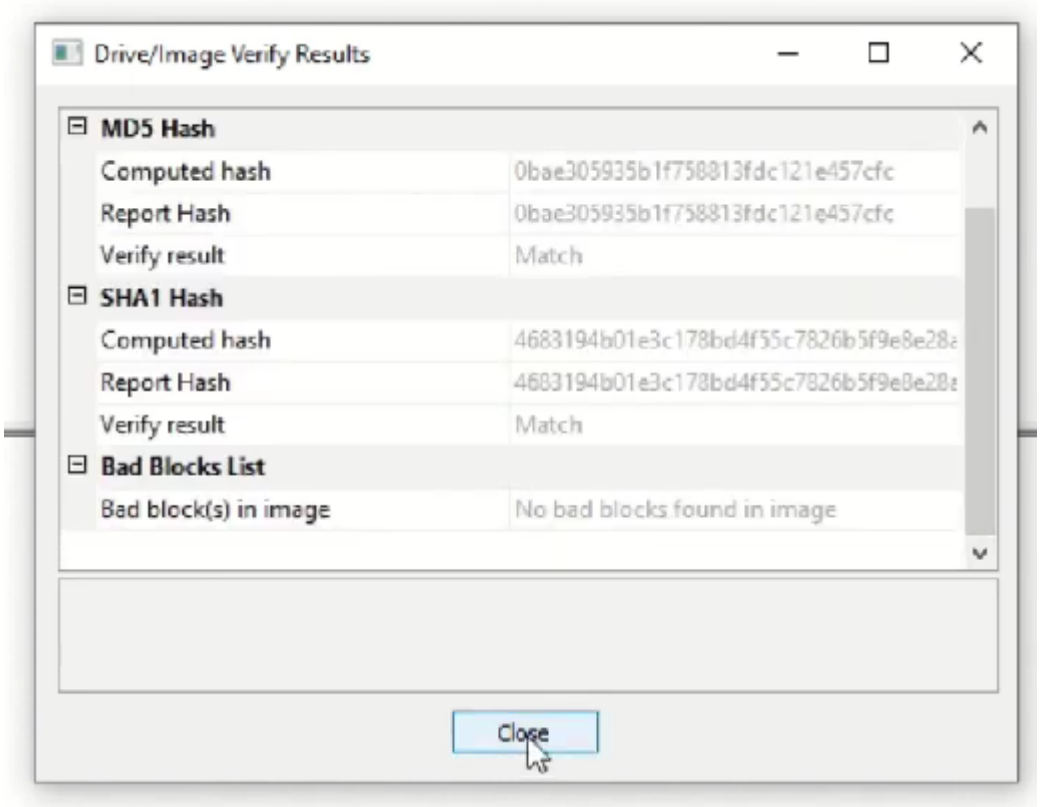
- **Fill the details of the evidence item**

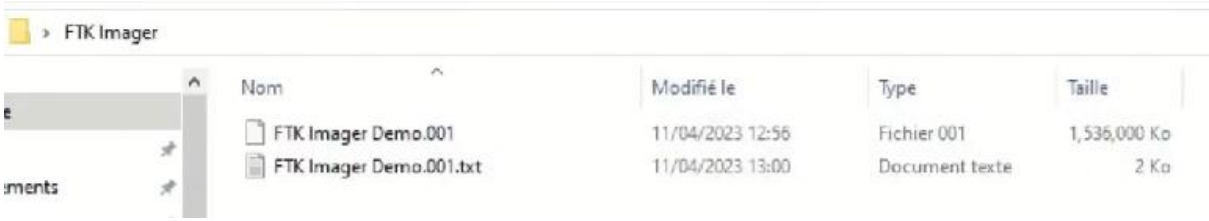


- Why: Imaging preserves original evidence; FTK is industry-standard.



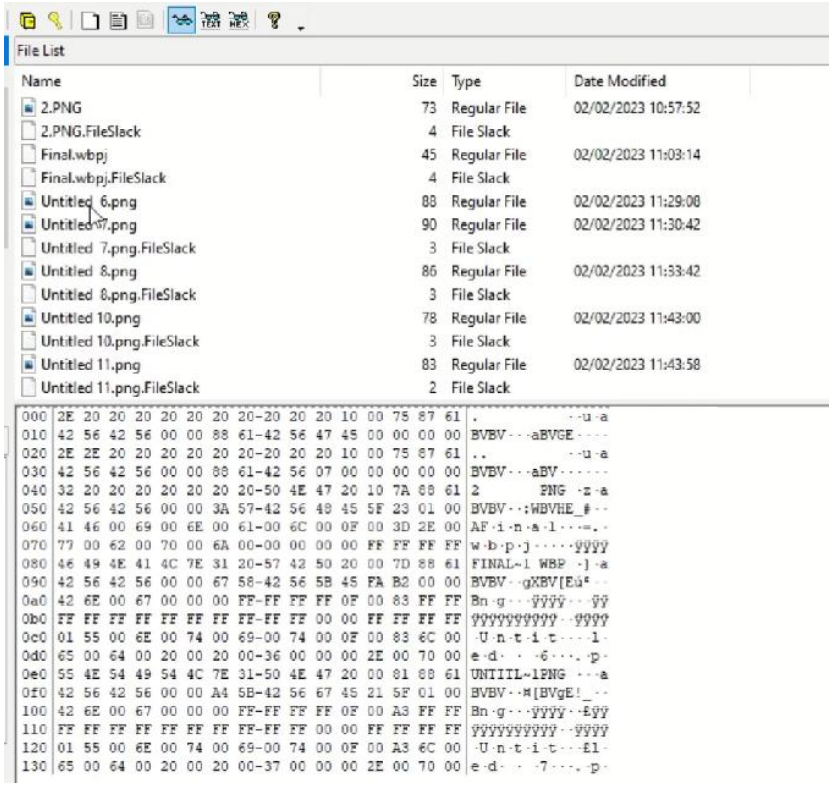
- **Image results**





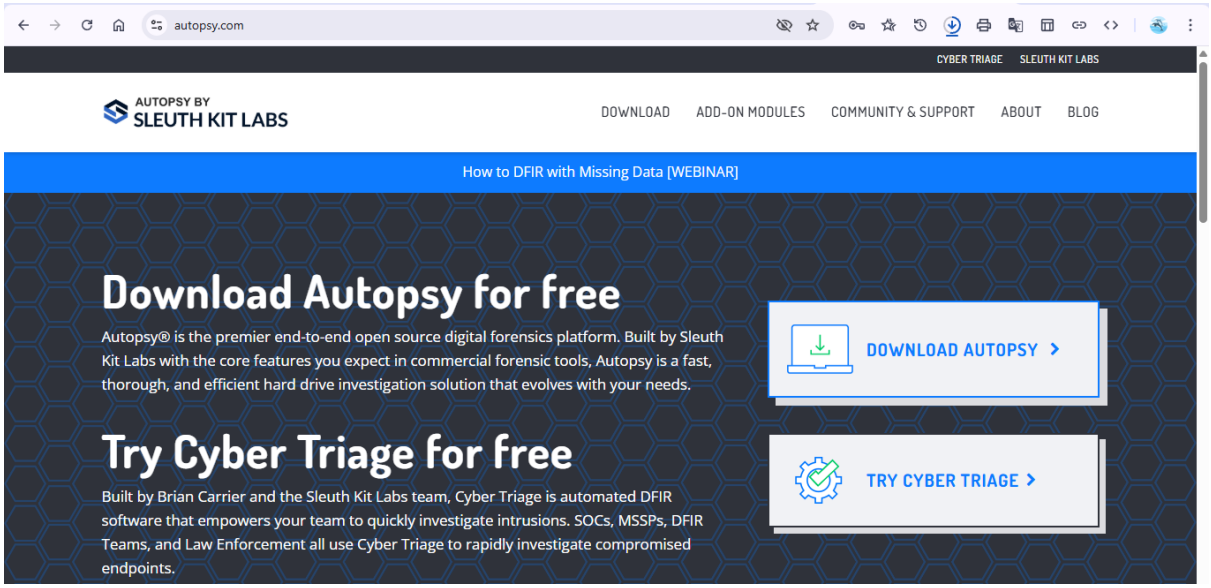
- **Results output**

```
Physical Evidentiary Item (Source) Information:
[Device Info]
 Source Type: Physical
[Drive Geometry]
 Cylinders: 485
 Tracks per Cylinder: 255
 Sectors per Track: 63
 Bytes per Sector: 512
 Sector Count: 7,802,880
[Physical Drive Information]
 Drive Model: Verbatim USB Device
 Drive Serial Number: AA00000000000485
 Drive Interface Type: USB
 Removable drive: True
 Source data size: 3810 MB
 Sector count:    7802880
[Computed Hashes]
 MD5 checksum:    0bae305935b1f758813fdc121e457cfc
 SHA1 checksum:   4683194b01e3c178bd4f55c7826b5f9e8e28a149
```

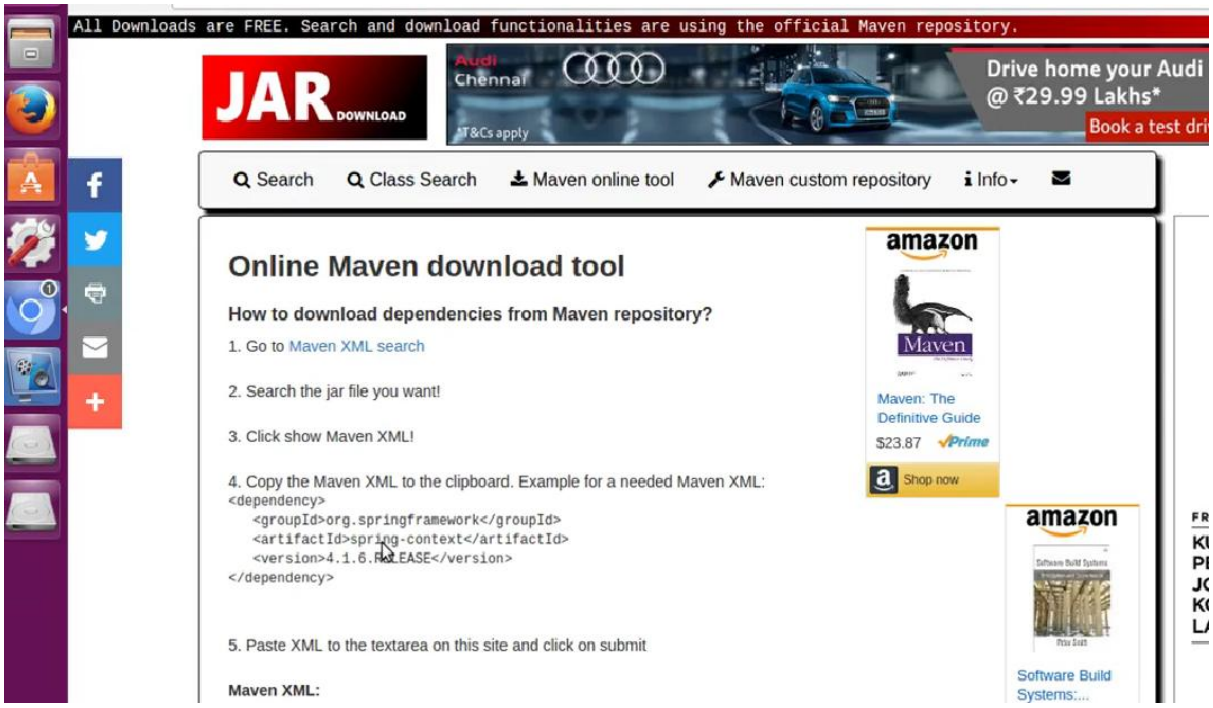


4. Configure Autopsy

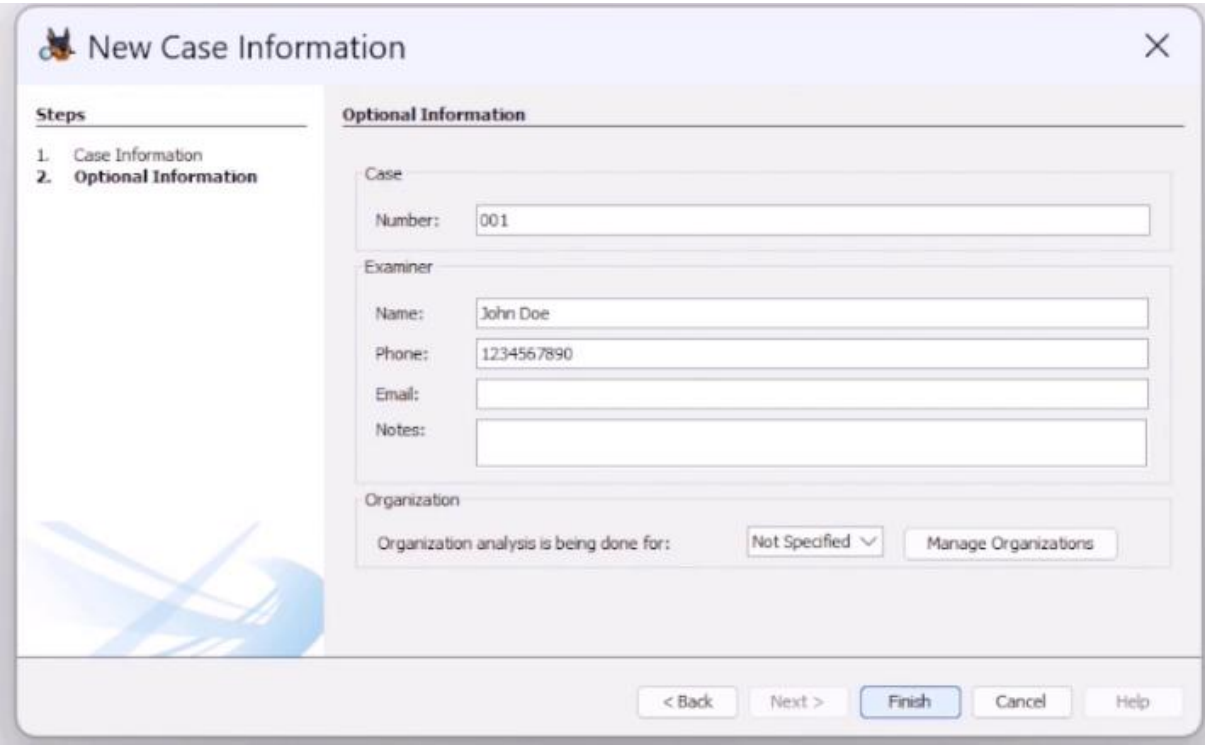
Steps:



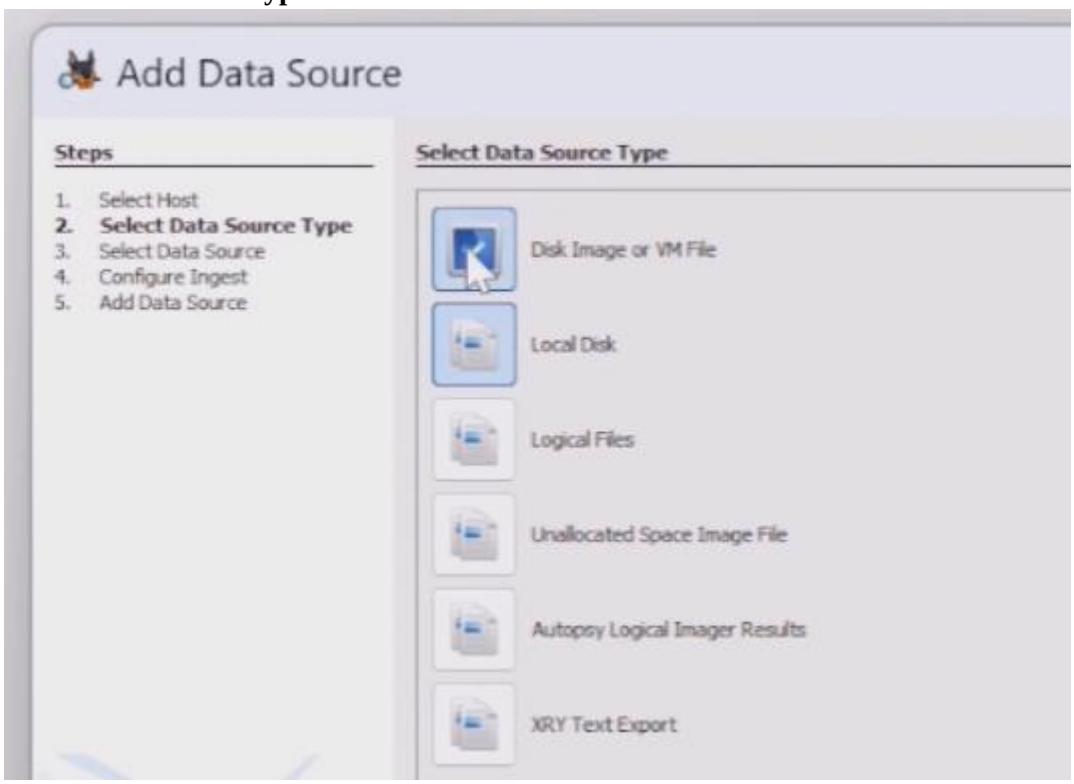
- Install Java and maven dependencies.

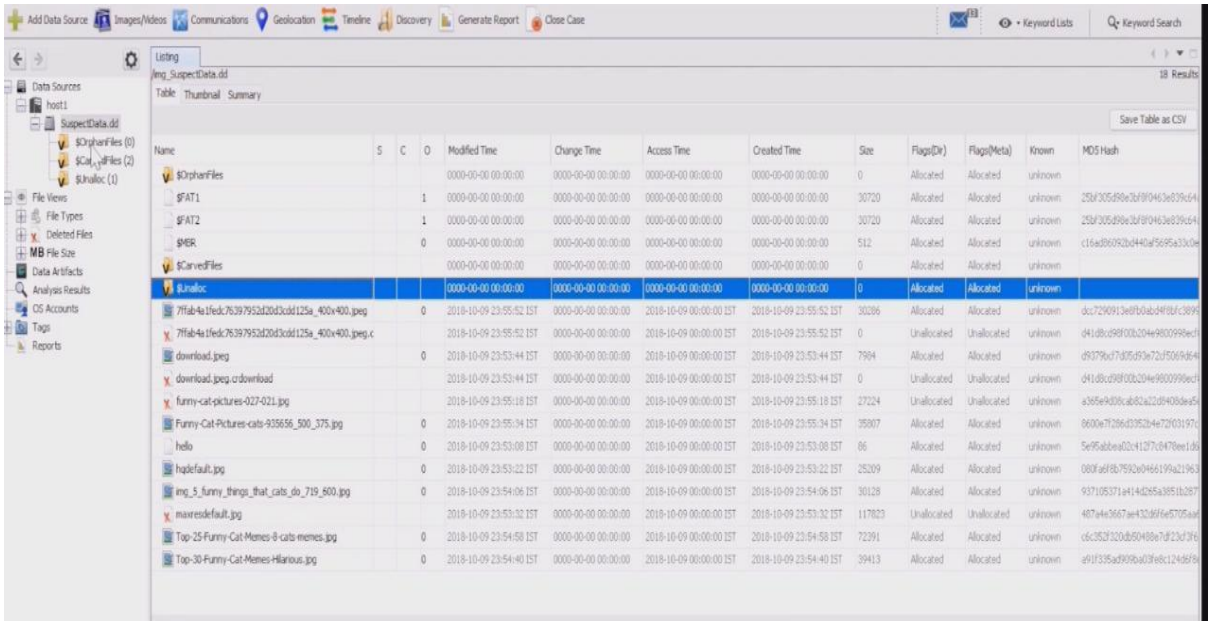


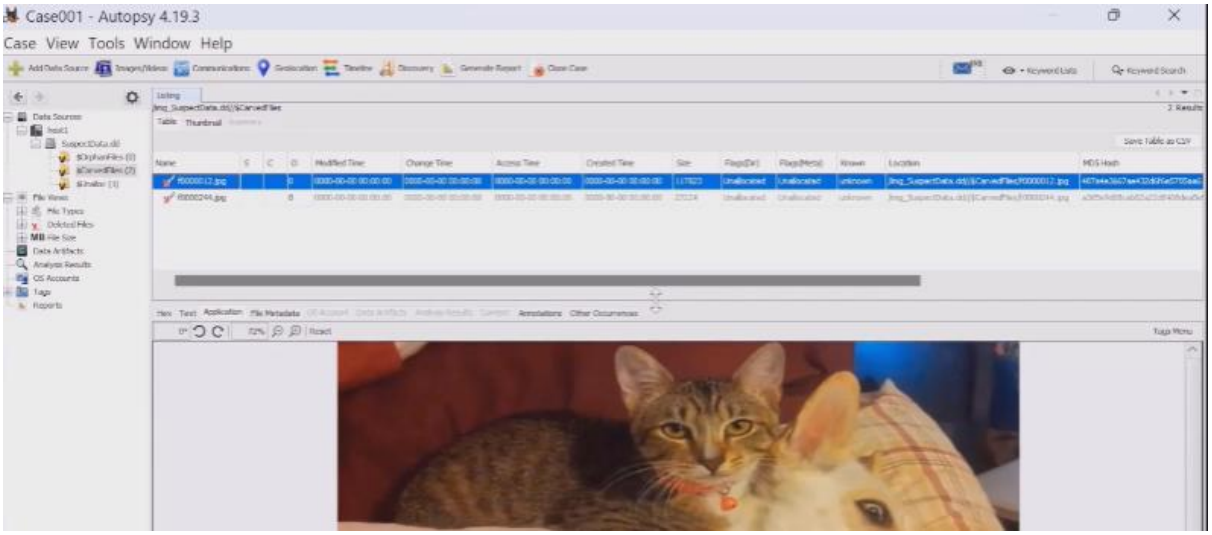
- Import a sample disk image and analyze partitions.

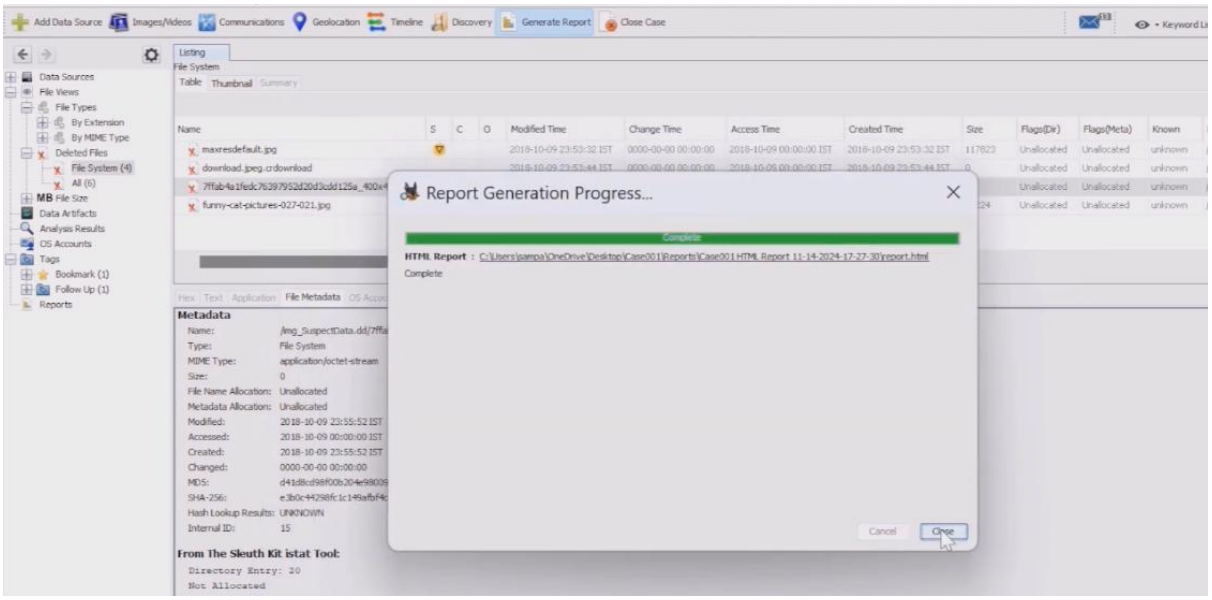


Select the file to encrypt

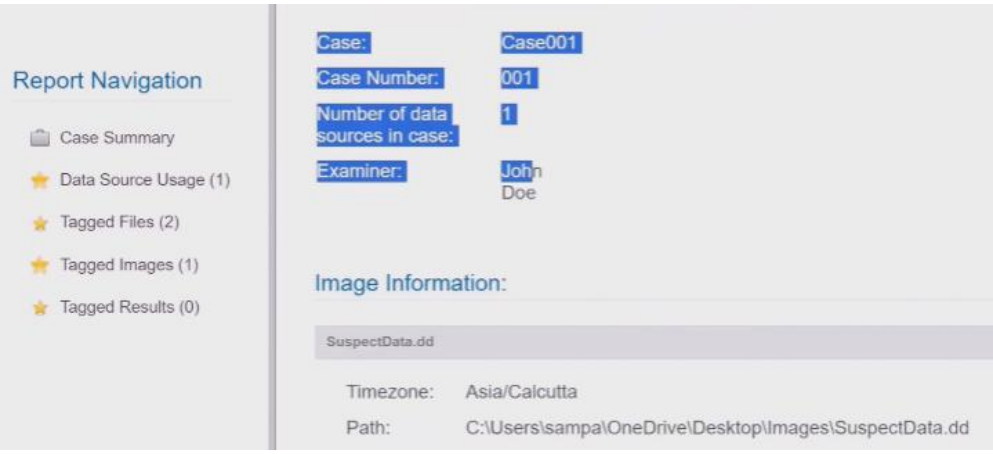


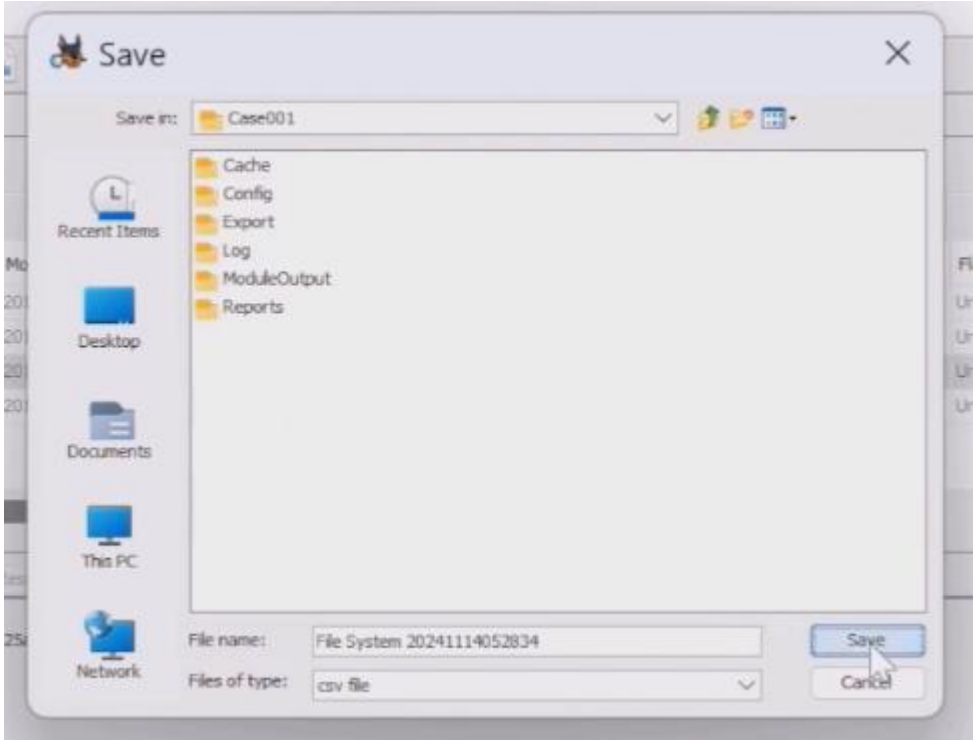






- Why: Autopsy automates timeline analysis, keyword searches, and file recovery.

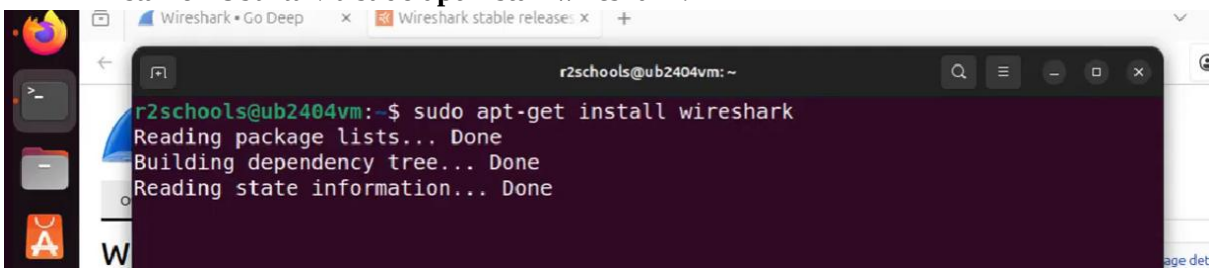


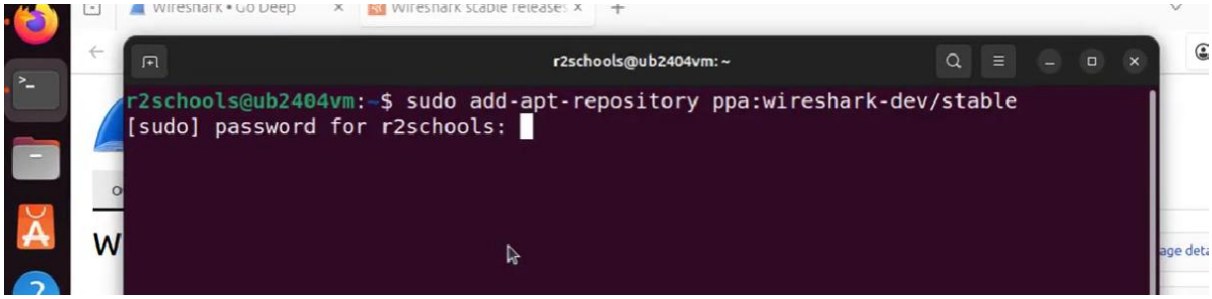


5. Setup Wireshark

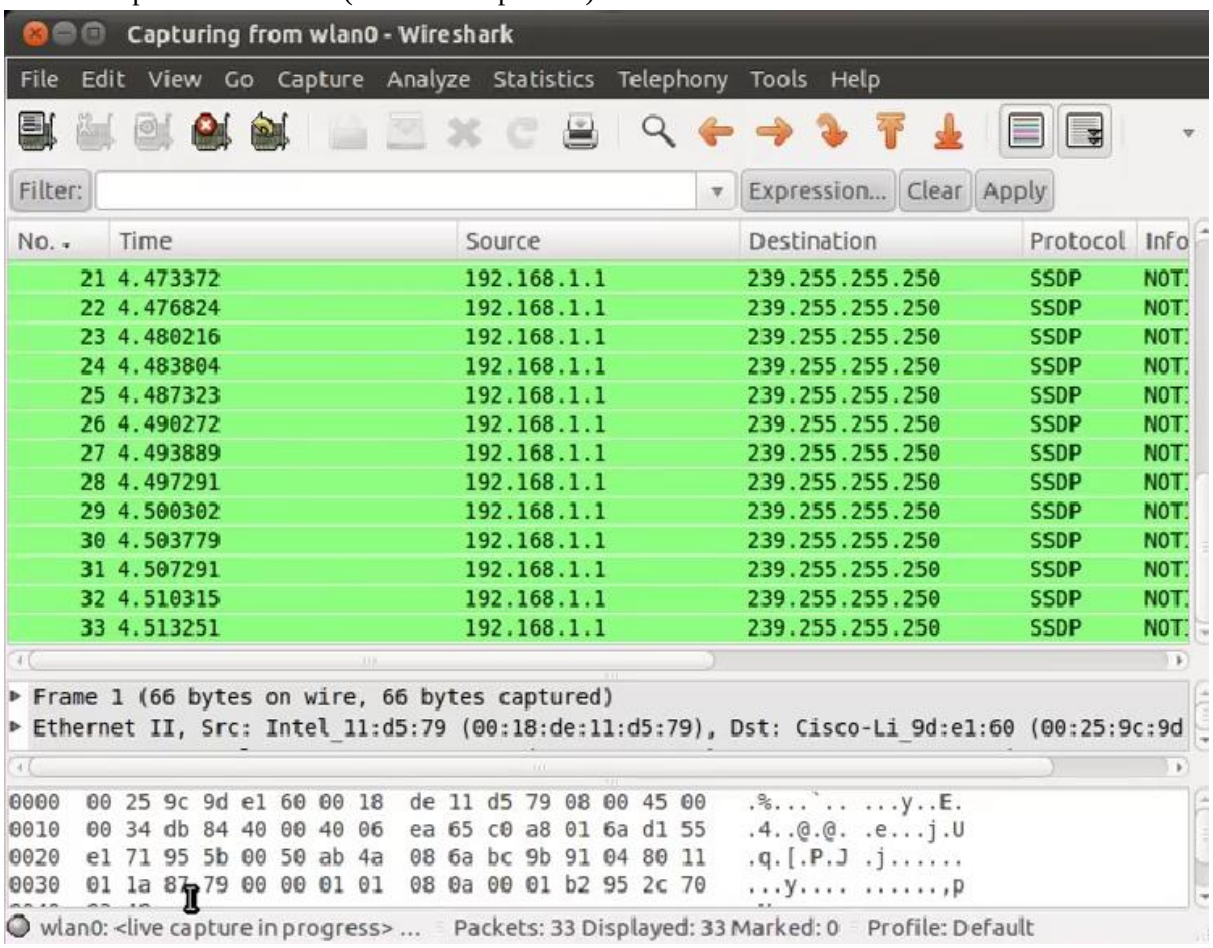
Steps:

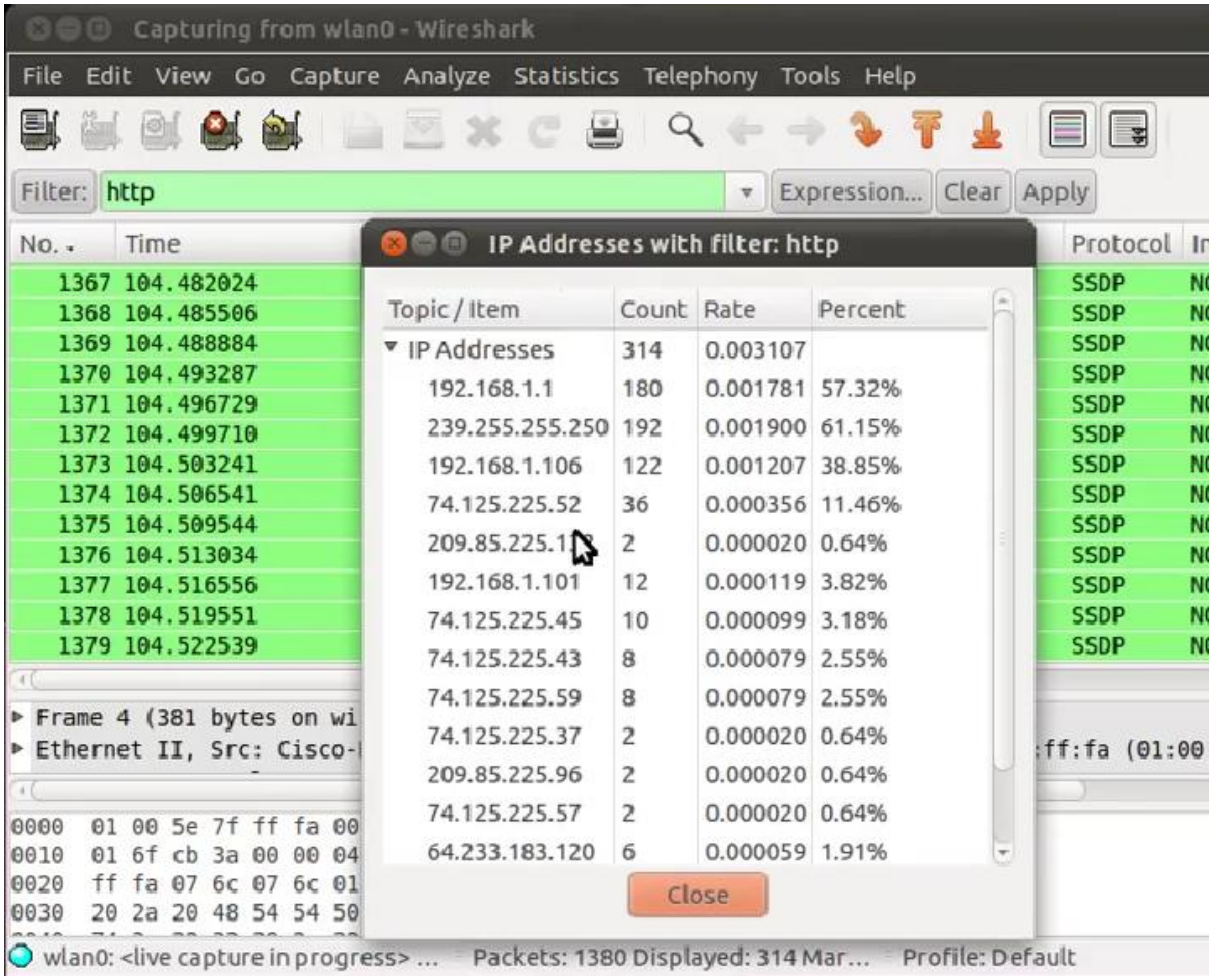
- Install on Ubuntu via **sudo apt install wireshark**.

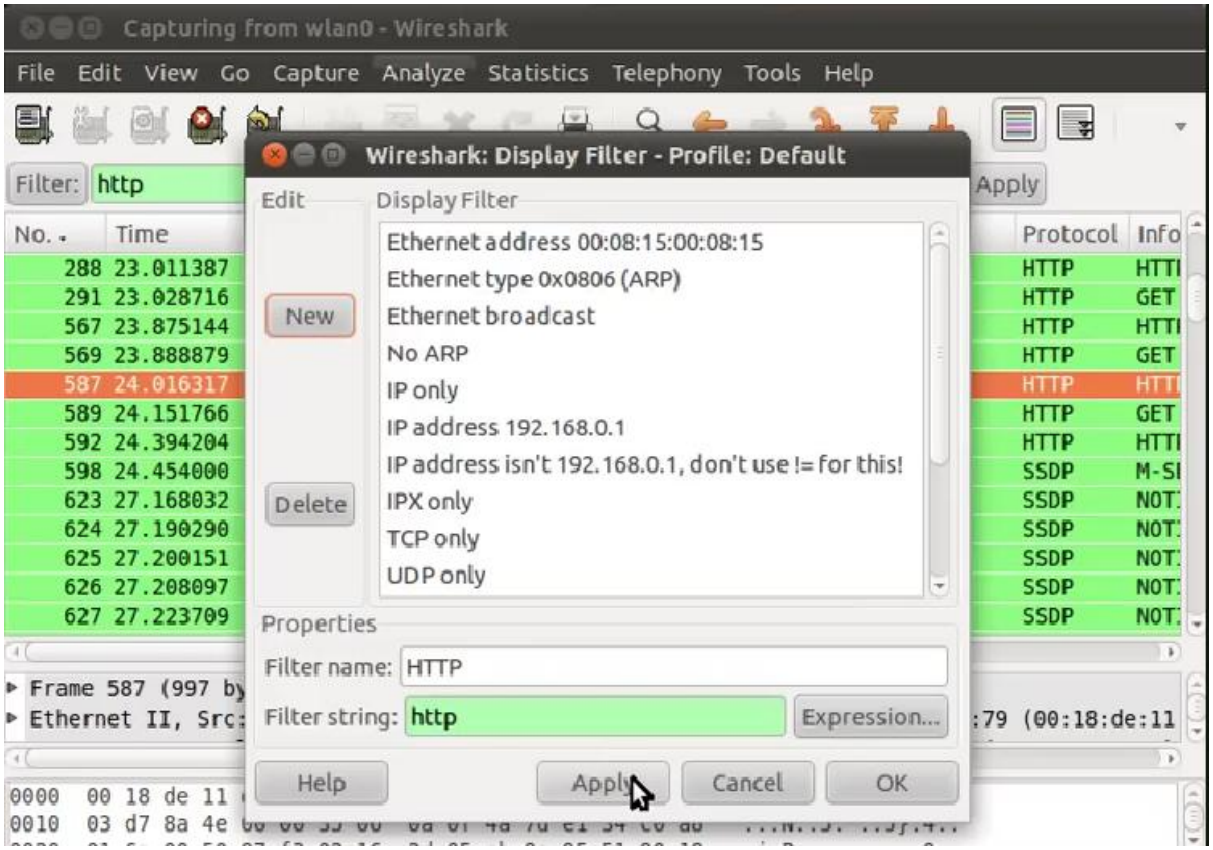




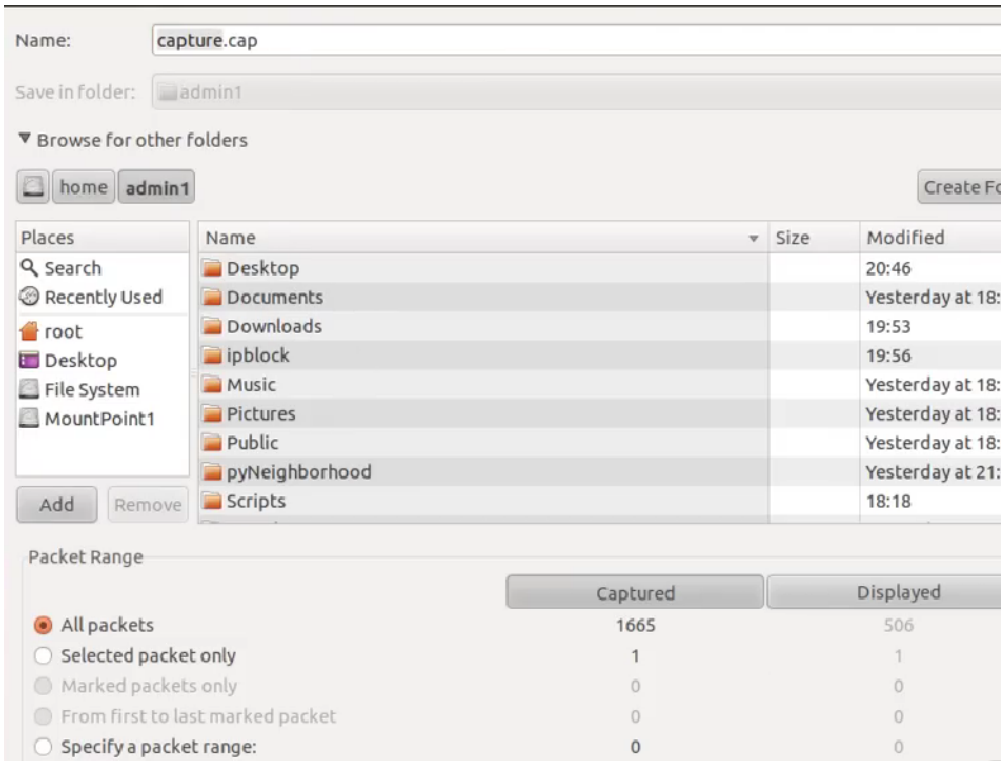
- Capture local traffic (filter HTTP packets).




Capturing from wlan0 - Wireshark
File Edit View Go Capture Analyze Statistics Telephony Tools Help
Filter: http
Expression... Clear Apply
No. Time
1367 104.482024
1368 104.485506
1369 104.488884
1370 104.493287
1371 104.496729
1372 104.499710
1373 104.503241
1374 104.506541
1375 104.509544
1376 104.513034
1377 104.516556
1378 104.519551
1379 104.522539
Protocol
SSDP
Frame 4 (381 bytes on wi
Ethernet II, Src: Cisco-
0000 01 00 5e 7f ff fa 00
0010 01 6f cb 3a 00 00 04
0020 ff fa 07 6c 07 6c 01
0030 20 2a 20 48 54 54 50
IP Addresses with filter: http
Topic / Item Count Rate Percent
IP Addresses 314 0.003107
192.168.1.1 180 0.001781 57.32%
239.255.255.250 192 0.001900 61.15%
192.168.1.106 122 0.001207 38.85%
74.125.225.52 36 0.000356 11.46%
209.85.225.1 2 0.000020 0.64%
192.168.1.101 12 0.000119 3.82%
74.125.225.45 10 0.000099 3.18%
74.125.225.43 8 0.000079 2.55%
74.125.225.59 8 0.000079 2.55%
74.125.225.37 2 0.000020 0.64%
209.85.225.96 2 0.000020 0.64%
74.125.225.57 2 0.000020 0.64%
64.233.183.120 6 0.000059 1.91%
Close
wlan0: <live capture in progress> ... Packets: 1380 Displayed: 314 Mar... Profile: Default



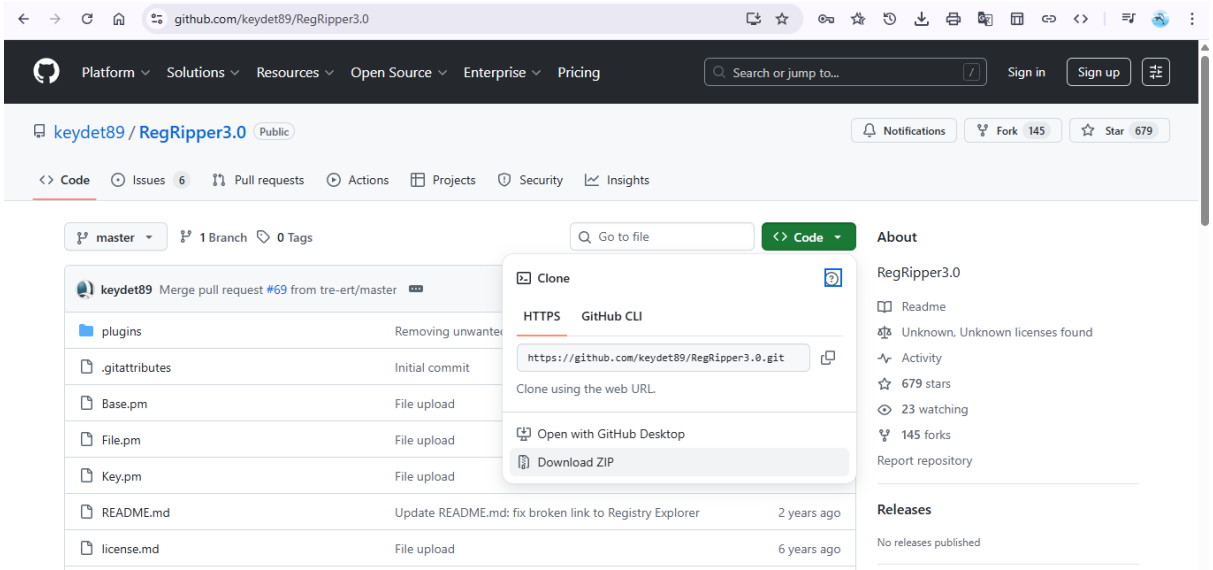
- Why: Network forensics helps trace intrusions or data exfiltration.



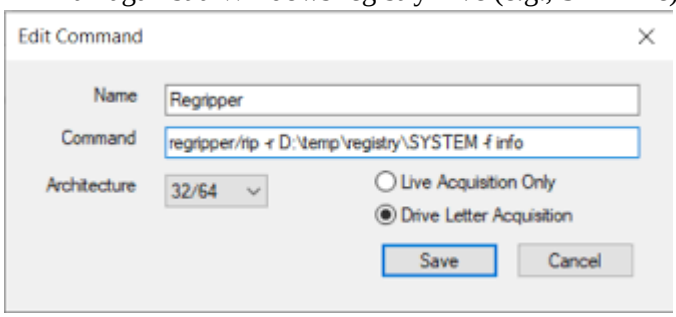
6. Install RegRipper

Steps:

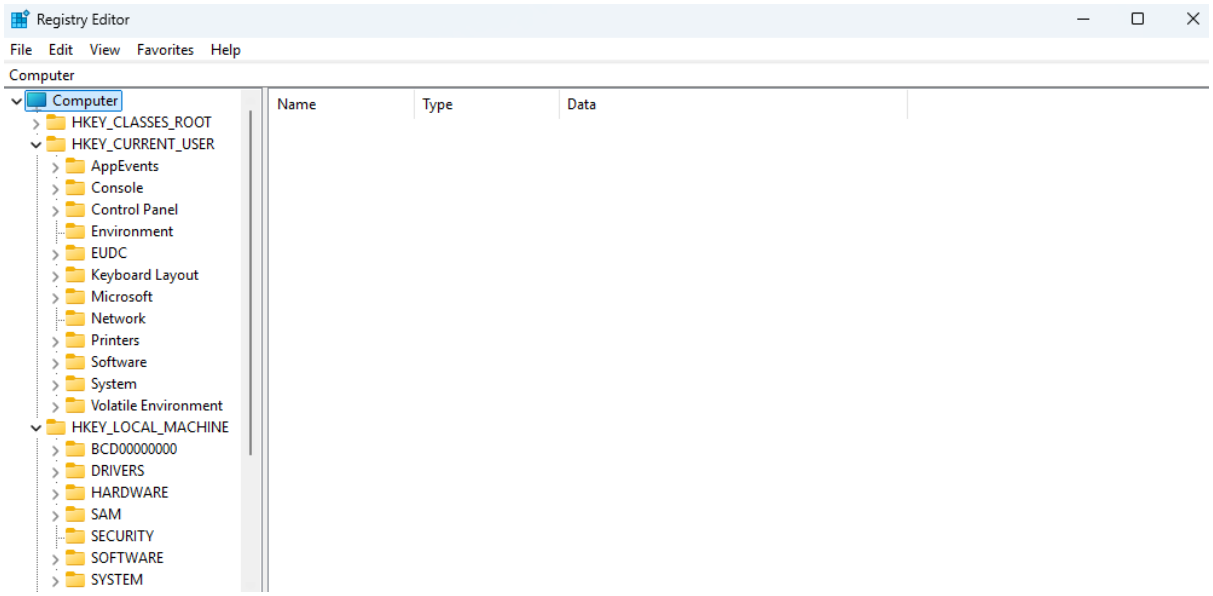
- Download from GitHub.



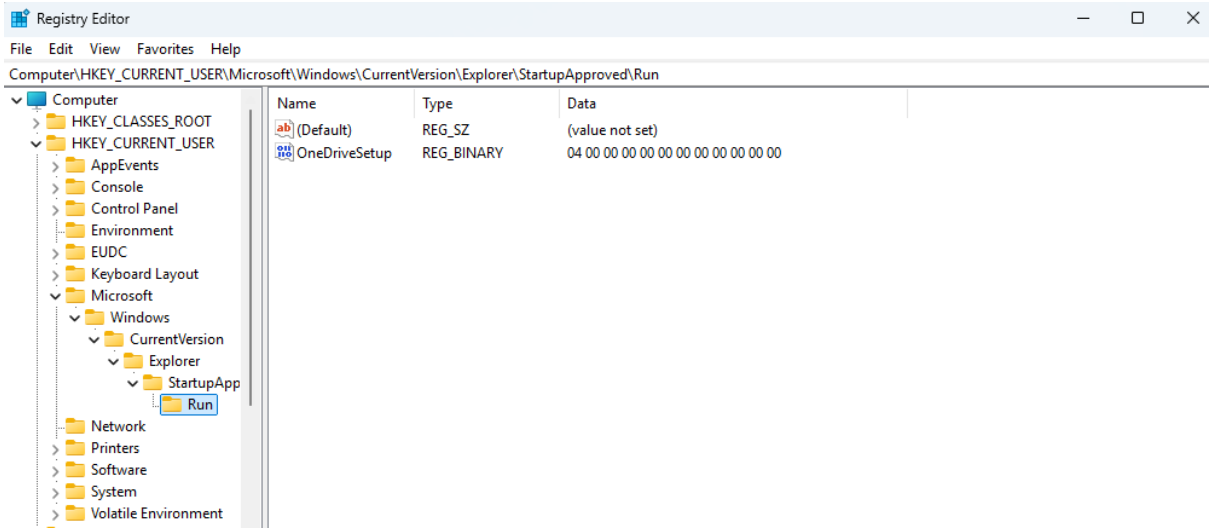
- Run against a Windows registry hive (e.g., SAM file).



- Why: Registry stores user activity, installed software, and system configs.



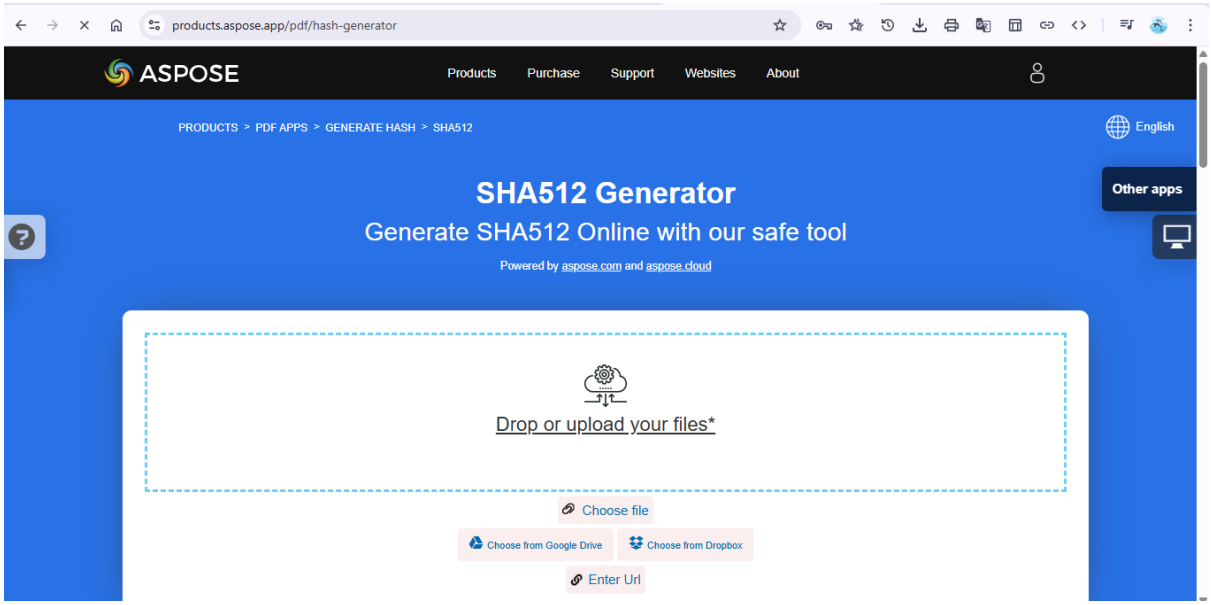
Results : Rootkeys, Keys, Values, Subkeys



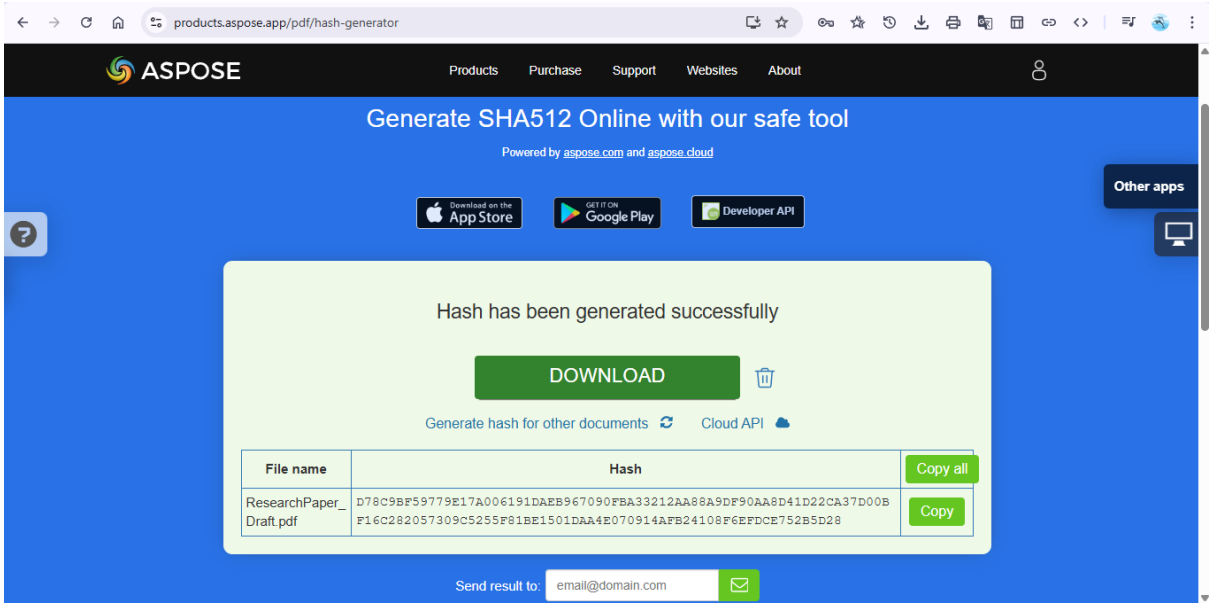
7. Configure HashMyFiles

Steps:

- Download the portable app

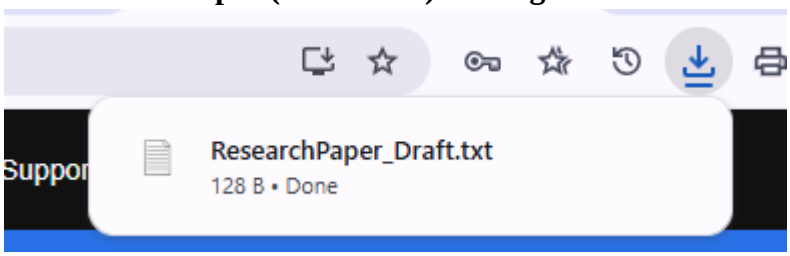


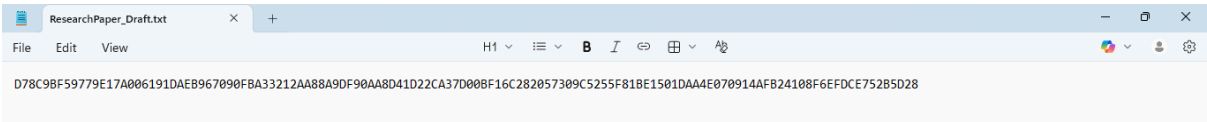
- Generate hashes for evidence files.



- Why: Hashes verify evidence integrity (e.g., match pre/post-analysis SHA-256).

Results and output (hash value) of the given file





Hash Value of the evidence file :
D78C9BF59779E17A006191DAEB967090FBA33212AA88A9DF90AA8D41D22CA37D00BF16C282057309C5255F81BE1501DAA4E070914AFB24108F6EFDCE752B5D28